



**TAKORADI
TECHNICAL
UNIVERSITY**

FACULTY OF ENGINEERING

DEPARTMENT OF OIL AND NATURAL GAS ENGINEERING

NATURAL GAS UTILISATION IN GHANA: DEVELOPMENT TRENDS, PROSPECTS AND CHALLENGES

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD
OF A BACHELOR OF TECHNOLOGY IN PETROLEUM ENGINEERING

BY

KEZIA QUARM

BC/BPR/20/009

Takoradi Technical University

August 2024.

DECLARATION

I hereby declare that this research project work titled “Natural Gas Utilization In Ghana: Development Trends, Prospects and Challenges” submitted as a requirement for the award of Bachelor of Technology in Petroleum Engineering at the Takoradi Technical University is entirely a result of hard work and research under the guidance of Mr. JJ Owusu. I am confident that this research project work has not been presented in any of the university. However, all sources have been duly acknowledged.

KEZIA QUARM,

BC/BPR/20/009.

DEDICATION

I dedicate this project work to God Almighty for bringing me this far. I also dedicate this project work to my parents for their explicit support and to my supervisor for his correct guidance.

ACKNOWLEDGEMENT

This undertaking report could not have been completed without the participation and assistance of so many people whose names may not all be enumerated. Their contributions are sincerely appreciated and gratefully acknowledged. However, I would like to express my deepest gratitude to my supervisor Mr. JJ Owusu and my family.

TABLE OF CONTENTS

NATURAL GAS UTILIZATION IN GHANA: DEVELOPMENT TRENDS, PROSPECTS AND CHALLENGES..... i

DECLARATION	ii
DEDICATION	iii
ACKNOWLEDGEMENT	iv
TABLE OF CONTENTS.....	v
ABSTRACT.....	vii
CHAPTER ONE: INTRODUCTION.....	1
1.1 Background to the Study.....	1
1.2 Statement of the Problem.....	1
1.3 Aim	2
1.4 Significance of Study.....	2
1.5 Scope.....	2
1.6 Limitations of Study	2
1.7 Organisation of Work	3
CHAPTER TWO: LITERATURE REVIEW.....	4
2.1 Natural Gas in Ghana.....	4
2.2 Properties of Natural Gas.....	5
2.3 Composition of Natural Gas	6
2.4 Forms of Natural Gas Utilisation in Ghana	7
2.4.1 Power Generation.....	7
2.4.2 Domestic and Industrial Use.....	7
2.4.3 Transportation	8
2.5 West African Gas Pipeline Project	8
2.6 Associated Problems with the WAGP	9
2.6.1 Hydrate Formation	9
2.6.2 Dissolution of Components.....	9
2.6.3 Climate Change.....	9
2.6.4 Flammability	10
2.7 Proposed Control Measures	10
2.7.1 Development of a Pragmatic Management Scheme	10
2.7.2 Development of a Predictive Model	10
2.8 Review of Related Works	10

CHAPTER THREE: DEVELOPMENT TRENDS OF NATURAL GAS UTILISATION IN GHANA	13
3.1 Data Source	13
3.2 Natural Gas Production and Import in Ghana.....	13
3.3 Pricing	14
3.4 Development of Natural Gas Utilisation in Ghana	15
3.4.1 Natural Gas Distribution and Transmission Pipeline Network.....	16
3.4.2 Natural Gas Demand and Supply.....	16
3.4.3 Liquefied Petroleum Gas Usage	17
CHAPTER FOUR: PROSPECTS AND CHALLENGES OF NATURAL GAS UTILISATION IN GHANA	19
4.1 Prospects for Natural Gas Utilisation in Ghana.....	19
4.1.1 Industrial Fuels.....	19
4.1.2 Gas for Power Generation.....	19
4.1.3 Transportation Gas.....	19
4.2 Challenges Facing Natural Gas Utilisation in Ghana	19
4.2.1 Industry Structure and Regulatory Framework.....	20
4.2.2 Gas Transmission and Distribution.....	20
4.2.3 Gas Storage Problems	21
CHAPTER FIVE: CONCLUSIONS AND RECOMMENDATIONS	22
5.1 Conclusions.....	22
5.2 Recommendations.....	22
REFERENCES	23

ABSTRACT

Although still relatively young, Ghana's natural gas sector carries substantial untapped potential that, if properly harnessed, could drive the country's next phase of sustainable economic growth. As a clean, cost-effective, and environmentally sound energy source, natural gas is increasingly favoured by energy users over conventional alternatives. This study examined the development trends, prospects, and challenges associated with natural gas utilization in Ghana. It relied on secondary data drawn from academic papers, journal articles, and published annual reports and outlooks issued by the Ghana National Gas Company, the Ghana National Petroleum Corporation, and the Energy Commission of Ghana.

Findings indicate that natural gas offers considerable promise across multiple sectors of the Ghanaian economy, whether deployed as an industrial feedstock, a transportation fuel, or a source for power generation. Domestic production has followed a steady upward trajectory and is projected to continue rising, while gas imports from Nigeria have become progressively less dependable. Despite these opportunities, the sector faces notable constraints, chiefly a shortage of gas gathering and distribution infrastructure, weaknesses in fiscal and regulatory policy, limited storage capacity, and insufficient investment. The study concludes that the absence of a conducive environment for private-sector participation, unfavourable gas pricing structures, and inadequate storage infrastructure to absorb the country's gas output collectively pose a serious threat to the sustainable development and effective utilization of Ghana's natural gas resources.

CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Demand for natural gas in Ghana has risen sharply in recent years, driven largely by its role as fuel for electricity generation in thermal plants, a trend linked to the 5.8% annual growth in national electricity consumption (Anon, 2010). Installed generation capacity has struggled to keep pace with this rapidly growing demand, leading to recurring power shortages and outages. Hydroelectric dams, which serve as the country's base-load source, already operate near maximum output, while thermal plants running on light crude oil remain an unreliable option given the volatility of global crude prices.

Natural gas presents a viable alternative to crude-oil-fired thermal generation, but the West African Gas Pipeline (WAGP), which has historically supplied Nigerian gas to these plants, has in recent years become increasingly unreliable and insufficient. In response, government and private-sector actors have pursued efforts to commercialize Ghana's own associated natural gas, previously flared off during oil production, as a substitute fuel source. To this end, a taskforce was established in 2011 to drive the monetization of domestically produced gas and curb flaring (Anon, 2015). A dedicated midstream gas company was subsequently created to develop the infrastructure needed to receive associated gas from upstream operations and process it into lean gas for thermal plants as well as liquefied petroleum gas (LPG) for household and vehicle use (Sulemana et al., 2017).

To support this, a 150 MMBtu/day gas processing plant and a 144-kilometre lean gas transmission pipeline were built alongside the existing WAGP infrastructure, enabling the processing and transport of both domestic and imported Nigerian gas to thermal facilities. Given the relatively modest volume of Ghana's gas reserves and the existing domestic demand from thermal power generation, the country's resources are not considered large enough to justify a large-scale Liquefied Natural Gas (LNG) export project (Suleman, 2020). Nonetheless, the sector's development into a mature, investment-ready industry continues to be hampered by several obstacles, chief among them a lack of upstream infrastructure and the ongoing difficulty of determining an industry structure suited to a still-emerging gas sector.

1.2 Statement of the Problem

The discovery of substantial oil and gas deposits in 2007, followed by the start of production in 2010, was widely expected to transform Ghana's energy landscape and stimulate economic growth through export revenue and expanded electricity generation capacity. With natural gas accounting for 37% of the country's total petroleum output as of 2016 (Anon, 2020), the resource represents a significant opportunity for national economic development, most immediately for the power sector, which has for years grappled with unreliable Nigerian gas supply under contracts dating back to 2009 (Occhiali and Hafner, 2017).

Pipeline-imported gas has grown steadily less dependable, owing both to feedstock shortages in Nigeria and to Ghana's own difficulties in meeting its debt obligations under the supply agreement. The result is a widening gap between rising domestic demand for natural gas and the quantity actually available. Closing this gap, and meeting growing demand going forward, makes it essential for Ghana to turn to its own gas reserves, pursuing their effective extraction and the productive use of their end products to help drive national economic growth.

As of 2017, Ghana held proven natural gas reserves of 0.80 trillion cubic feet (Tcf), placing it 72nd globally and representing roughly 0.012% of the world's total proven reserves of 6,923 Tcf. At a daily production rate of approximately 80 Mscf/d against annual consumption of 43,502 Mcf, the country's proven reserves are equivalent to about 18.4 times its yearly consumption. At current consumption levels and excluding any unproven reserves, this implies roughly eighteen years of supply remaining (Anon, 2017a). More than a decade after these discoveries, however, comparatively little progress has been made toward effectively exploiting and utilizing this substantial resource, with the sector continuing to operate at a loss even as gas prices remain high (Anon, 2021a). This study therefore sets out to examine the development trends, prospects, and challenges surrounding the effective utilization of Ghana's natural gas.

1.3 Aim

To investigate the development trends, prospects and challenges of natural gas utilization in Ghana and find ways to make natural gas a more valuable resource for the economy.

1.3.1 Objectives

1. To study the development trends of natural gas use in Ghana and its potential.
2. To identify the challenges facing natural gas use in Ghana.
3. To find solutions to address these challenges and improve natural gas use in Ghana.

1.4 Significance of the Study

The significance of this project is to understand the utilization of natural gas across various sectors while addressing the challenges involved, and to provide insights that support optimizing natural gas utilization for Ghana's sustainable development.

1.5 Scope

This study is limited to assessing the current state of natural gas utilization in Ghana, analyzing development trends, and evaluating the prospects and challenges faced across various sectors, along with recommendations to address them.

1.6 Limitations of the Study

1. The study's findings are limited to the available data and may not be fully generalizable across all sectors, transportation, power generation, industrial, and domestic, in Ghana.
2. No survey was conducted; instead, the study relies on a review of existing publications and information from relevant institutions.

1.7 Organisation of the Study

Chapter 1 provides the general introduction, covering the background of study, statement of the problem, objectives, significance, scope, and limitations. Chapter 2 presents a review of relevant literature. Chapter 3 presents the data sources and development trends. Chapter 4 examines the prospects and challenges of natural gas utilization in Ghana. Chapter 5 presents the conclusions and recommendations.

CHAPTER TWO: LITERATURE REVIEW

2.1 Natural Gas in Ghana

Among fossil fuels, natural gas stands out as the cleanest and most economical option for long-term power generation, serving both industrial and commercial users in support of economic growth. It ranks among the world's principal energy sources, meeting roughly one-fifth of global energy demand, compared with about a third supplied by crude oil and a quarter by coal. Its environmental profile, clean combustion, and versatility across applications (Table 2.1) make it especially significant for Ghana's power sector, which accounts for the bulk of domestic natural gas consumption (Jacobs, 2011).

Table 2.1

Difference Between Oil and Gas (Source: Agbonifio, 2013)

Particular	Crude Oil	Natural Gas
Energy Density	High	Low
Storage	Inexpensive/easy	Expensive/difficult
Transport	Inexpensive/easy	Expensive/difficult
Infrastructure	No need for pressure vessel	All pressurized/cryogenic
Market	Fungible	Tied to dictated infrastructure
Environment	More pollution/ Higher CO ₂	Lower CO ₂ /clean burning

Ghana's total natural gas reserves stand at 6.4 trillion cubic feet (Tcf) (Anon, 2012), comprising 2.2 Tcf of associated gas and 4.2 Tcf of non-associated gas. Against daily production of 80 Mscf/d and annual consumption of 43,502 Mcf, the country's proven reserves amount to roughly 18.4 times current yearly consumption, equivalent to about eighteen years of supply at present usage rates, excluding any unproven reserves (Anon, 2017a).

Ghana's upstream sector has discovered and continues to produce from several fields, notably the Jubilee Fields, the Greater Jubilee Fields, the Tweneboa-Enyenra-Ntomme (TEN) Fields, and the Sankofa Gas Project. Gas from the Jubilee Fields, once flared off as a by-product of crude oil extraction, is now consumed domestically following the construction of midstream processing infrastructure. With an estimated 13.9 bcm of associated gas in the Jubilee fields, 10.3 bcm in the

TEN fields, and 31.4 bcm of non-associated gas in the Sankofa field, the sector represents a considerable asset for Ghana's economic development (Occhiali and Hafner, 2017). Given the relatively modest scale of these reserves and existing domestic demand from thermal power generation, however, they are not considered sufficient to support a large-scale Liquefied Natural Gas (LNG) export venture (Suleman, 2020). The sector's evolution into a mature, investment-ready industry nonetheless continues to face obstacles, including limited upstream infrastructure and unresolved questions around the most suitable industry structure for a still-developing market.

Despite these challenges, Ghana's gas sector remains in its early stages, leaving considerable room for growth toward sustained economic development. While gas imports have grown increasingly unreliable, a consequence of both feedstock shortages in Nigeria and Ghana's difficulties meeting debt obligations tied to the West African Gas Pipeline (WAGP), domestic market development has likewise been constrained by several structural barriers. These include an underdeveloped local market, weak infrastructure, the absence of a coherent gas policy framework and fiscal incentives, technical limitations, and the remoteness of key gas resources, all of which pose fundamental constraints on the sector's future development (Suleman, 2020).

2.2 Properties of Natural Gas

Key physical and chemical properties of natural gas include boiling point, freezing point, specific gravity, density, flammability limits, ignition temperature, carbon and hydrogen content, and relative density, each relevant to safe handling, pipeline design, and combustion performance. Natural gas composition and properties can vary by production field and may be adjusted seasonally by distributors, making accurate characterization important for both technical and economic purposes.

Figure 2.1

Physical and Chemical Properties of Natural Gas (Source: Eswara, 2013)

Property	Value	Remarks
Boiling Point	$-161.5^{\circ}C$	@ 1 atm
Freezing Point	$-182.6^{\circ}C$	@ 1 atm
LNG Specific Gravity	$0.43 \sim 0.47$	rel to water = 1
Gas density	$0.7 \sim 0.9$	kg/m^3 @ STP
Flammability limits	$4 \sim 15$	by volume in air
Ignition Temperature	$538^{\circ}C$	@ 1 atm
Carbon content	73	by weight
Hydrogen content	24	by weight
Oxygen content	0.4	by weight
Hydrogen/Carbon atomic ratio	$3.0 \sim 4.0$	
Relative Density	$0.72 \sim 0.8$	at $15^{\circ}C$
Octane Number	$120 \sim 130$	
Methane Number	$69 \sim 99$	

2.3 Composition of Natural Gas

Natural gas is a naturally occurring combustible mixture of hydrocarbon gases. While natural gas is composed primarily of methane, it can also include ethane, propane, and non-hydrocarbons such as hydrogen sulphide and carbon dioxide. The composition of natural gas can vary widely; Table 2.2 below delineates the typical makeup of natural gas before it is refined.

Table 2.2

Typical Composition of Natural Gas (Source: Anon, 2013)

Component	Formula	Typical Range
Methane	CH ₄	70-90%
Ethane	C ₂ H ₆	0-20%
Propane	C ₃ H ₈	0-20%
Butane	C ₄ H ₁₀	0-20%
Carbon Dioxide	CO ₂	0-8%
Oxygen	O ₂	0-0.2%
Nitrogen	N ₂	0-5%
Hydrogen Sulphide	H ₂ S	0-5%
Rare gases	Ar, He, Ne, Xe	Trace

At its purest, natural gas is almost entirely methane, a simple molecule of one carbon atom bonded to four hydrogen atoms (CH₄). Because natural gas is naturally odourless, the compound mercaptan is added before it reaches end-users, giving it its characteristic “rotten egg” smell and allowing leaks to be detected by odour alone. Sulphur occurs in natural gas primarily as hydrogen sulphide (H₂S); once its concentration exceeds 5.7 mmg per m³, the gas is classified as “sour.” Sour gas, containing both H₂S and CO₂, becomes highly corrosive and acidic in the presence of moisture, and because H₂S is toxic and hazardous, sour gas must undergo sweetening before use. The hydrocarbons commonly associated with methane, including ethane and propane, follow the general formula C_nH_{2n+2} and belong to the alkane family. Gas is described as “dry” when it consists almost entirely of methane with other hydrocarbons largely removed, and as “wet” when those additional hydrocarbons remain present (Anon, 2013). The working ratio for real-life applications is defined by Equation 2.1 (Olivier, 2013):

$$Wr = \frac{LHV}{\sqrt{d} T} \quad (2.1)$$

Within Ghana, gas composition varies by field in terms of API gravity and sulphur content. Gas from the Sankofa field, produced via the John Agyekum Kufuor FPSO vessel, is a medium-density, low-sulphur product with an API gravity of 28.9 and sulphur content of 0.30 wt% (Vitol, 2017). Gas from the Tweneboa-Enyenra-Ntomme (TEN) field, known as the TEN Blend, carries a higher API gravity of 36.25 and sulphur content of 0.39 wt% (Anon, 2020). Gas from Ghana’s

various fields is generally refinery-friendly, meaning it requires little additional sulphur-reduction processing, which enhances its market value (Anon, 2017b).

2.4 Forms of Natural Gas Utilisation in Ghana

Both imported and domestically produced natural gas in Ghana serve several end uses, with the largest share directed to power generation and the remainder split between domestic consumption and transportation.

2.4.1 Power Generation

Ghana's electricity generation capacity currently falls short of demand, resulting in recurring power disruptions and load-shedding. This shortfall carries serious implications for the country's economic prospects. Historically, Ghana's economic development has been constrained by unreliable and costly power generation fuel, compounded by weak institutional capacity and a difficult business environment (Fritsch, 2015). While hydropower has long been the nation's dominant electricity source, its relative share has declined in recent years as thermal generation fuelled by natural gas has grown in importance, a shift closely tied to the 2007 discovery of domestic oil and gas resources (Occhiali and Hafner, 2017).

Ghana's electricity is generated from a mix of hydroelectric dams and thermal plants running on crude oil, natural gas, or diesel. Of the country's ten thermal plants, which together supply 60% of national electricity, five operate on dual fuel (oil or gas), two run exclusively on natural gas, and three on crude oil (Anon, 2020). In 2017, total gas supplied to thermal plants climbed to roughly 43 million Btu, nearly 60% above 2016 levels, with only about 17% arriving via the WAGP from Nigeria and the remaining 73% sourced domestically from the Atuabo gas processing plant operated by Ghana Gas (Anon, 2017b).

The West African Gas Pipeline terminates in Ghana at the Volta River Authority plant in Takoradi. The Tema-Takoradi pipeline segment, with a capacity of $1.70 \times 10^9 \text{ m}^3$, is designed to meet the Takoradi Power Plant's demand of $1.22 \times 10^9 \text{ m}^3$, alongside supplying the Effasu hydropower facility, and is also expected to serve the VALCO plant in Tema, which requires $0.60 \times 10^9 \text{ m}^3$. The pipeline's offshore segment runs submerged at depths between 26 and 70 metres (Waheed, 2009).

As is typical elsewhere, demand from the power sector has been the principal driver behind the growth of Ghana's domestic gas market. Given that natural gas remains cheaper than oil-based alternatives and already fuels more than 70% of the country's thermal capacity, its share of total gas demand is projected to remain as high as 80% through 2040 (Occhiali and Hafner, 2017).

2.4.2 Domestic and Industrial Use

Liquefied Petroleum Gas (LPG) has largely displaced kerosene and charcoal as the fuel of choice for household cooking in Ghana, becoming a near-universal feature of homes and food vendors alike. In many developed economies, by contrast, LPG-equivalent gas is piped directly into homes. On the industrial side, natural gas serves multiple purposes: as a fuel for process heating, in combined heat-and-power systems, as feedstock for producing chemicals, fertilizers, and hydrogen, and as fuel for plant and lease operations (Anon, 2020a).

2.4.3 Transportation

Compressed natural gas (CNG) offers a cleaner alternative to petrol and diesel for vehicles. In Ghana, however, the trend has moved in the opposite direction, with many vehicle owners shifting away from CNG back toward conventional fuels, largely due to the higher relative cost of petrol per mile compared to CNG, despite CNG's advantages for both the environment and human health. Natural gas also fuels aviation in the form of liquefied natural gas (LNG), though this application remains largely undeveloped in Ghana at present. Within the transportation sector more broadly, natural gas is additionally used to power the compressors that move gas through pipeline networks (Anon, 2008).

2.5 West African Gas Pipeline Project

Nigeria's Niger Delta holds approximately 78% of Africa's natural gas reserves, with the remainder distributed across Libya, Algeria, Egypt, Angola, Mozambique, Namibia, Tunisia, and Tanzania. Much of Nigeria's gas remains underutilized and is instead flared into the atmosphere, generating heat, smoke, and particulate pollution while imposing a considerable economic cost on the country. Flaring in the region is estimated at 56.5×10^6 Mcf per day, representing roughly a quarter of global gas flaring (Obanijesu and Sonibare, 2005).

To help curb this flaring and develop Nigeria's substantial gas fields, the World Bank recommended the West African Gas Pipeline (WAGP) project. The pipeline was designed to deliver 450 MMscf of Nigerian gas to power generators and industrial users across Ghana, Benin, and Togo, via 617 km of new offshore pipeline and 57 km of onshore pipeline connected to an offtake point on the existing 359 km Escravos-Lagos gas system. Estimated at \$550 million, the project encompassed pipeline construction, metering, pressure regulation, gas scrubbing, and compression facilities, and was developed through a joint venture involving NNPC, ChevronTexaco West African Gas Pipeline Co. Ltd, Shell Overseas Holdings Ltd, and Takoradi Power Co. Ltd (Obanijesu, 2009).

Construction on this cross-border project began in 2005, with the pipeline designed for a service life of at least 20 years before requiring renewal. A mid-line pressure relief system sits along the Otta-Iddi Iroko expressway in Ogun State as part of the onshore network. An 18,000-horsepower compressor station at Lagos Beach connects to a 20-inch pipeline running 51 km onshore before extending a further 15 km offshore and turning west across the Nigeria-Benin border toward Cotonou. From there, the pipeline diameter narrows to 16 inches for 267 km to Tema, then to 12 inches for another 233 km to Takoradi, before a final 12-inch, 80 km offshore segment connects Takoradi to Effasu in Ghana (Obanijesu, 2009).

The pipeline's Ghanaian endpoint is the Volta River Authority plant in Takoradi. The Tema-Takoradi segment, built at an initial cost of \$106 million, was designed to deliver 1.70×10^9 m³ of gas to meet the Takoradi Power Plant's 1.22×10^9 m³ demand, alongside supplying a nearby hydropower plant at Effasu, and was expected to also serve the VALCO plant in Tema, with its 0.60×10^9 m³ requirement. The offshore section runs submerged at depths of 26 to 70 metres (Waheed, 2009).

Pipeline transport remains the standard method for moving natural gas worldwide. Shell Transmission, for instance, owns or holds interests in 17 natural gas pipelines across the Gulf of

Mexico, with combined capacity nearing 9 billion cubic feet per day (Anon, 2004). As established networks continue operating successfully, more countries have pursued similar cross-border infrastructure: in 2005, Saipem began building a 370 km, 48-inch gas export pipeline between Qatar and the UAE, targeted for completion in 2006 (Saipem, 2004), while Libya similarly began constructing gas pipelines to Egypt and Tunisia in 2004 (Buddy, 2004). Nigeria's pipeline network, however, has drawn significant public and regulatory scrutiny due to a comparatively high rate of pipeline failures, which occur when the pipe body fails as a result of metallurgical or manufacturing defects (Obanijesu et al., 2006). Beyond vandalism, the WAGP faces substantial risk from corrosion, faulty welding, operational errors, defective piping, and equipment malfunction (Obanijesu et al., 2006). When failures occur in the offshore section, transmitted gas escapes directly into the surrounding marine environment, threatening local ecosystems. Each such incident produces a strong vertical flow of gas from the leak point to the surface, known as vertical dispersal, the behaviour of which is shaped largely by prevailing sea currents (Saipem, 2004).

2.6 Associated Problems with the WAGP

Operation of the WAGP pipeline network is affected by several recurring challenges that reduce its overall efficiency and reliability. Key issues include the following.

2.6.1 Hydrate Formation

Because natural gas boils at -162°C , it forms hydrates when released into water. These crystalline solids develop when gas components interact with water under conditions of high pressure and low temperature. Hydrate formation, which sometimes accompanies the vertical movement of escaped gas, can produce effects ranging from behavioural disturbances to chronic toxicity in exposed organisms, depending on the volume of gas involved and duration of exposure (Patin, 2004).

2.6.2 Dissolution of Components

Untreated natural gas typically contains small but significant quantities of undesirable compounds such as carbon dioxide (CO_2), hydrogen sulphide (H_2S), and oxygen, all of which pose risks to nearby communities and aquatic life. When released into water, these compounds can dissolve into acidic byproducts that marine organisms may absorb, potentially entering the human food chain through bioaccumulation and posing health risks. Acidified water can also be taken up by aquatic plants through their root systems, restricting their growth. Dissolved CO_2 further affects seawater temperature, salinity, ice cover, turbulence, and current patterns, abiotic changes that in turn carry broader ecological consequences.

2.6.3 Climate Change

Nigerian natural gas typically consists of 60-90% methane, which does not readily dissolve in water. Under normal conditions, much of this gas would be expected to oxidize via bacterial action before reaching the surface; however, the pressure at which gas escapes from a failure point often prevents this from occurring. Methane has an estimated global mean atmospheric lifetime of roughly 8.4 years (Prather et al., 1995), with a perturbation lifetime of around 12 years and the ability to travel 500-1,000 km per day via lower-atmosphere wind transport.

As a potent greenhouse gas, methane contributes significantly to climate change. Rising atmospheric methane concentrations through the late twentieth century prompted international efforts toward annual reduction targets over the past two decades. In recognition of its atmospheric burden and climatic impact, the Kyoto Protocol classified methane among the greenhouse gases requiring control. Any inadvertent release of gas through WAGP failures therefore works directly against these global reduction goals, and has raised broader concerns about the potential impact of such releases on marine ecosystems (Obanijesu, 2009).

2.6.4 Flammability

Because natural gas is lighter than air, any gas escaping from an underwater pipeline failure will rise through the water column and disperse into the atmosphere, where it can ignite or burn in the presence of an ignition source, potentially causing fire damage across a wide surrounding area.

2.7 Proposed Control Measures

Given the range of risks associated with potential failures anywhere along the pipeline's length, the following control measures are recommended.

2.7.1 Development of a Pragmatic Management Scheme

Pipeline operators should establish a comprehensive management strategy that includes provisions for compensating affected individuals or communities in the event of an incident. Rapid response capability, including prompt deployment of emergency response personnel, should be a central component. This requires well-trained staff equipped with the necessary resources, alongside a regularly updated inventory (ideally reviewed quarterly) to ensure prompt replacement of used materials. Given the offshore nature of the pipeline, helicopter access should also be factored into response planning, and appropriate insurance coverage should be arranged for all personnel involved (Obanijesu, 2009).

2.7.2 Development of a Predictive Model

Since pipeline failure risk can be reduced but never fully eliminated, there is a clear need for proactive, comprehensive risk assessment to identify pollution-prone areas and prepare spill contingency plans. Central to this effort would be mathematical models capable of predicting how leaked gas travels through water, built around its four primary transport mechanisms: vertical bubble transport, vertical turbulent diffusion, component dissolution, and horizontal advection driven by ambient currents. Such models should be run using appropriate numerical simulation techniques that account for both upstream and downstream effects, with the resulting algorithms implemented in software tools that operators can use to forecast the spread of leaked gas over time (Obanijesu, 2009).

2.8 Review of Related Works

Anon (2013) examined Ghana's natural gas sector across three dimensions: demand and supply dynamics, institutional challenges facing the Ghana National Gas Company (GNGC) and Bulk Oil Storage and Transportation (BOST), and the upstream oil and gas industry. The study found that Ghana had been slow to invest in gas supply and power generation infrastructure, and that an unattractive gas pricing policy had discouraged private-sector investment in the sector. It further

identified inefficiencies in the operations of state-owned energy agencies. However, the study offered limited new insight into the petroleum midstream segment specifically.

Anon (2010) took a broader view, surveying natural gas demand and its various utilization pathways, including power generation, fertilizer and methanol production, and industrial, commercial, and transportation use, while also addressing related aspects of the petroleum value chain, including supply-demand balances, production scenarios, and gas pricing projections. The study focused particularly on the development of gas infrastructure, notably the West African Gas Pipeline (WAGP) and the Jubilee Gas Project, and concluded with recommendations centred on petroleum industry regulation, covering marketing arrangements, ownership structures, and legal frameworks. It fell short, however, of offering a holistic assessment of the industry's prospects and challenges. In a related vein, Anon (2012) investigated natural gas demand, supply, pricing policy, and the associated institutional and regulatory frameworks, finding that as of 2012 gas price policy centred largely on pricing and marketing arrangements. The study identified two core regulatory pillars: first, that the Ghana National Gas Company would serve as the aggregator responsible for trading, processing, transporting, and storing natural gas; and second, that Bulk Oil Storage and Transportation would oversee gas transmission infrastructure connecting suppliers to end users.

Adams (2009) contributed further insight, identifying key upstream development challenges relating to investment levels, petroleum economics, and the absence of clear legal and fiscal frameworks. He recommended that Ghana engage international oil companies to secure both capital and technology, and proposed the establishment of an independent Petroleum Regulatory Authority to separate the Ghana National Petroleum Corporation's regulatory function from its role as an industry participant. In a later study, Adam (2014) examined petroleum revenue management, finding that the government had by then commercialized petroleum revenues worth approximately two billion US dollars, and concluding that this economic strategy was aimed at improving fiscal terms and supporting exploration and development activity.

Among the earlier significant contributions to the field, Amponsah and Opei (2014) analysed Ghana's downstream petroleum industry and its principal actors, identifying a range of operational challenges including weak infrastructure, limited access to capital, currency depreciation, underperformance at the Tema Oil Refinery (TOR), poor maintenance practices, and government delays in settling under-recovery payments. While the authors recommended restructuring BOST and TOR toward greater self-sufficiency, they did not specify how such restructuring might be implemented in practice. Their further recommendation, that the private sector be encouraged to increase petroleum product imports, is, in this writer's assessment, difficult to reconcile with the challenges the study itself identified. Increased private importation does little to address underlying infrastructure deficiencies, and arguably risks compounding rather than resolving them.

More recently, Sulemana (2020) examined the restructuring of Ghana's natural gas industry with a focus on value optimization, applying the concept of vertical integration to assess the sector's institutional structure. The study engaged with an ongoing debate among industry stakeholders over the most appropriate structural model for Ghana's gas industry, concluding that the current arrangement is comparatively inefficient and requires reform. It recommended preserving

GNPC's aggregating role, separating infrastructure ownership from upstream gas producers, and assigning an independent entity, such as BOST, to operate the GNGC transmission pipeline.

CHAPTER THREE: DEVELOPMENT TRENDS OF NATURAL GAS UTILISATION IN GHANA

3.1 Data Source

This study used existing information from previous research, books, articles, reports, and policy documents. Data sources included academic journals, government agencies (Ghana National Petroleum Corporation and Ghana Energy Commission), and international organisations (including the International Energy Agency).

3.2 Natural Gas Production and Import in Ghana

Ghana's upstream sector comprises several major projects, including Jubilee, TEN, Sankofa (OCTP), and HESS, with combined proved recoverable natural gas reserves expected to be produced over a period of roughly two to three decades based on current infrastructure and production schedules. The domestic gas supply chain is overseen by the Ghana National Petroleum Corporation (GNPC), which holds a mandatory equity stake in oil and gas agreements entered into within the country. Domestic natural gas production began with associated gas from the Jubilee field, followed by production from the TEN field, and production has risen substantially since, aided by non-associated gas from the Sankofa (OCTP) development. Imported gas arrives via the West African Gas Pipeline (WAGP), while domestically produced gas is handled by the Ghana National Gas Company (GNGC).

Figure 3.1

WAGP Gas Flows (Fulwood and Bros, 2018; Anon, 2018a)

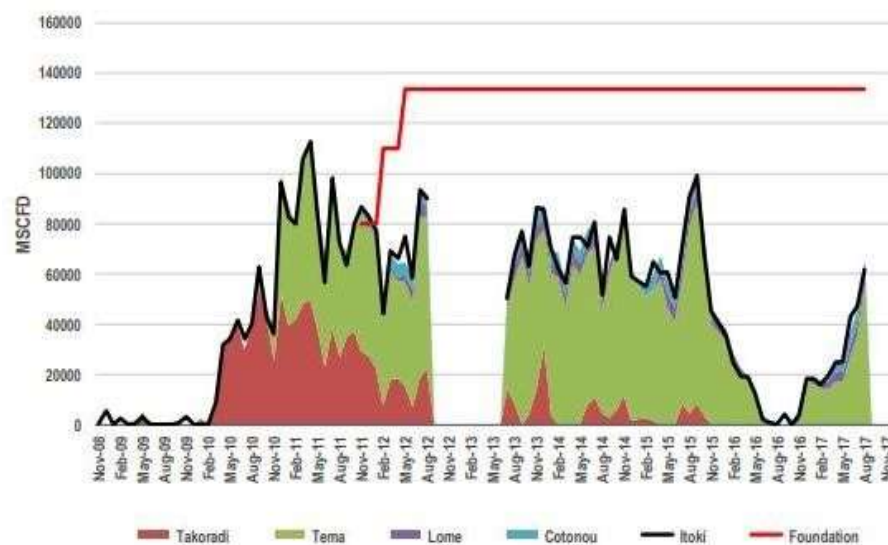
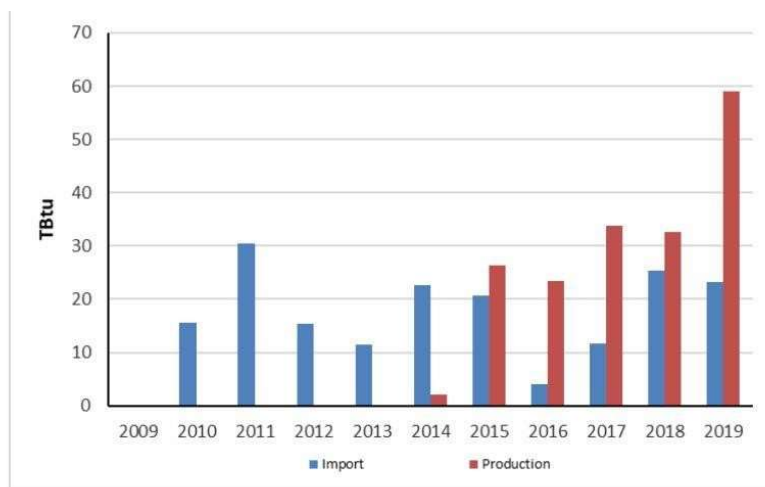


Figure 3.2

Trend in Natural Gas Production and Import (Anon, 2020a; Anon, 2020b)



3.3 Pricing

Prior to the start of production at the Jubilee field, Ghana depended entirely on imported gas, and prices were effectively set by the terms of the WAGP supply contracts. The price of this imported gas comprises three core components governed by those contracts:

1. The contracted wellhead price in Nigeria, made up of a base price that adjusts partly in line with Bonny Light crude oil prices and US inflation;
2. The transport tariff on the Escravos-Lagos pipeline system, which moves gas from the producing fields to the WAGP's entry point at Itoki; and
3. The WAGP tariff itself, structurally intended as a fixed capacity charge but, due to repeated force majeure declarations by the Nigerian supplier, applied in practice on a volumetric basis.

Additional fees and levies are layered on top of these to arrive at the final delivered price. Since WAGP operations began in 2011, gas delivered to Ghana, whether at Tema or Takoradi, has consistently priced in the mid-\$8MMBtu range. Over time, the WAGP tariff component has risen (now exceeding \$5/MMBtu), even as the wellhead price has declined alongside falling oil prices. At current Nigerian oil price levels, the wellhead price for gas destined for the WAGP is only about half what producers could earn selling the same gas domestically to Nigerian power generators or industrial users, an imbalance that incentivises producers to favour the domestic Nigerian market over WAGP exports. The combination of a high WAGP tariff and the resulting total delivered cost has also discouraged new customers from entering the market (Fulwood and Bros, 2018).

Following the commissioning of the pipeline and processing infrastructure needed to commercialise Jubilee’s associated gas in 2014, this gas was initially routed to power units in Takoradi. The first 200 billion cubic feet were supplied free of charge, with roughly a quarter of that volume delivered by mid-2017.

By 2016, GNPC was charging Takoradi power plants \$8.84/MMBtu for this gas, broken down into a commodity price of \$2.90/MMBtu (linked to light crude oil prices), a gathering, processing, and transportation fee of \$5.28/MMBtu, and a Public Utilities Regulatory Commission (PURC) levy of \$0.66/MMBtu. Notably, this final price closely mirrored the cost of imported WAGP gas, despite the underlying gas itself being supplied at no cost to GNPC, suggesting that the pricing structure allowed GNPC to generate substantial margins through the associated charges. Gas from the TEN fields, which yields both associated and non-associated streams, carries a commodity price that varies depending on the gas type (Fulwood and Bros, 2018).

Table 3.2

Gas Price Comparison at Tema (\$/MMBtu) (Abudu, 2019)

Charge	Nigeria 2016H1	Nigeria 2016H2	Jubilee	TEN Associated	TEN Non- Associated	Sankofa
Wellhead price	1.4870	1.2582	2.9000	0.5000	3.0000	6.6000
Nigeria Transport	1.2983	1.2083	-	-	-	-
Ghana Transport	-	-	5.2800	5.2800	5.2800	1.0000
WAGP transport	5.0330	5.0330	-	-	-	1.5000
Other charges	0.3158	0.3673	0.6600	0.6600	0.6600	-
Total	8.1341	7.9578	8.8400	6.4400	8.9400	9.1000

Beyond these core components, delivered gas prices also incorporate a regulatory fee, a pipeline protection zone charge, a delivery fee, a fuel-use charge, and a fixed credit support cost. As of the period under review, end-user gas prices in Ghana’s power sector had settled within the \$8-\$9/MMBtu range, closely tracking the cost of WAGP gas imported from Nigeria. At this pricing level, GNPC appears to be earning considerable profit margins on domestic gas sales.

3.4 Development of Natural Gas Utilisation in Ghana

How effectively a nation utilizes its natural resources depends largely on the extent of their development. Globally, natural gas development has expanded steadily over time, driven chiefly by its relatively clean environmental profile and lower cost compared to crude oil. Since past trends often signal future direction, assessing natural gas development trends is essential to

forecasting the resource's future economic value and sustainability for Ghana. Key areas of natural gas development include gas demand and supply, the distribution and transmission pipeline network, and the growth of LPG usage (Anon, 2008).

3.4.1 Natural Gas Distribution and Transmission Pipeline Network

Natural gas is a special commodity. The construction of its transmission and distribution pipe network is essential to its delivery and utilisation by end users. With its receiving end at Aboadze Thermal Plant in Takoradi, the offshore West African Gas Pipeline became the first gas pipeline system laid in Ghana, in 2006. Several gas pipeline network systems have been laid since then, with the most recent reported in 2019, when Ghana Gas Company set up and laid pipes to power the relocated Karpowership from Tema to the Sekondi Naval Base. From 2006 to 2019, natural gas pipelines grew at an average annual rate of 9.26% in Ghana, which was lower than the average annual natural gas utilisation rate during the same period (Anon, 2020a). This indicates that significant development in natural gas use necessitates expansion of natural gas pipelines; it is important to keep natural gas transmission and distribution networks growing at a steady pace, as this is an important guarantee for the rapid growth of natural gas utilisation.

3.4.2 Natural Gas Demand and Supply

In the history of petroleum consumption, Ghana has been a net importer of crude oil, petroleum products, and natural gas. Over the years, natural gas supply has never reached demand expectations; as of 2016, total supply from import and local production was 164 MMcfd, compared to total demand of 300 MMcfd (Anon, 2016). These statistics suggest a 136 MMcfd deficiency, that is, about 45% of natural gas demand unmet. The demand for natural gas continues to rise, with power sector demand anticipated to rise substantially as new plant installations were added. Other industries requiring close to 230 MMcfd include the petrochemical and fertiliser industries, with others requiring around 60 MMcfd more. These demands total around 1,790 MMcfd of natural gas on average, a level most likely not to be met by supply from 2019 and beyond (Anon, 2019).

The Ghana Energy Commission reported that natural gas delivered through WAGP to Ghana between 2016 and 2018 averaged only 60 MMcfd, with delivery being inconsistent (Table 3.1). A detailed look at the import data reveals that from September 2012 to June 2013, there was no available natural gas supply; during this period the WAGP pipeline was damaged close to the Togo border, and was later repaired in July 2013. As a result, domestic natural gas output and imports combined reached a little below 900 MMcfd in 2019 (Table 3.1). Jubilee (104 MMcfd), TEN (50 MMcfd), Sankofa (180 MMcfd), and HESS (50 MMcfd) fields provided the expected supplies. 2020 also saw fairly steady gas output, with local production totalling 228 MMcfd and imports of 560 MMcfd, giving a total gas output for the year of 888 MMcfd.

Table 3.1

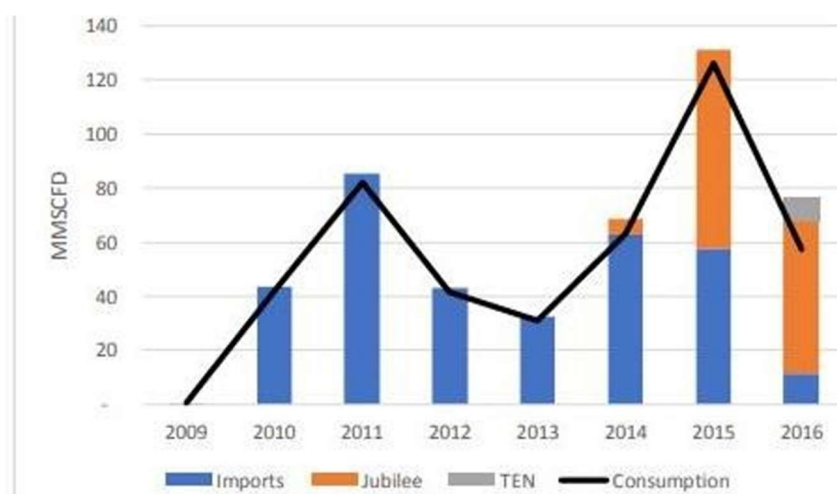
Total Daily Gas Supply, 2014 to 2020 (MMcfd) (Anon, 2020b)

Year	Project	Gas Production	Gas Import	Total Gas
2014	Jubilee	0	60	60
2015	Jubilee & TEN	83	60	143

Year	Project	Gas Production	Gas Import	Total Gas
2016	Jubilee, TEN & Sankofa	104	60	164
2017	Jubilee, TEN & Sankofa	104	60	164
2018	Jubilee, TEN & Sankofa	334	60	394
2019	Jubilee, TEN & Sankofa	334	560	894
2020	Jubilee, TEN & Sankofa	328	560	888

Figure 3.3

Historic Natural Gas Demand and Supply (Fulwood and Bros, 2018)



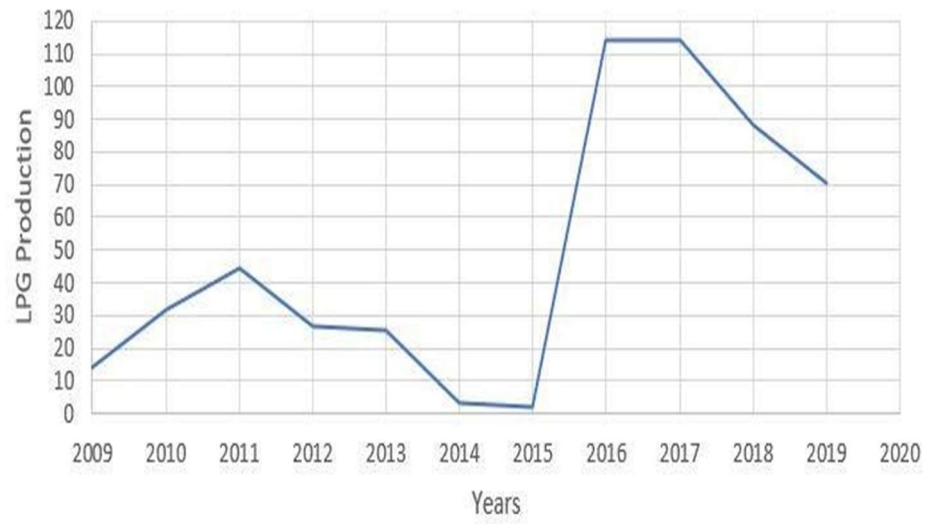
3.4.3 Liquefied Petroleum Gas Usage

Liquefied Petroleum Gas (LPG), predominantly made up of propane and butane, both constituents of natural gas, has over the years become a widely used household fuel in Ghana. About 60% of LPG stocks have historically been separated from raw gas and oil during extraction, with the remaining 40% a byproduct of crude oil refining (Anon, 2020a). This growth in LPG use has contributed to a rise in overall natural gas consumption and utilisation. In Ghana, LPG is used for a variety of residential purposes, most prevalently cooking, and government policy has historically aimed to expand LPG access nationally.

LPG production in Ghana saw fluctuation between 2009 and 2014. 2015 recorded the lowest production, at around 2 kilotons, while the following year saw a sharp increase to around 114 kilotons. 2017 recorded the highest production, at around 114.6 kilotons. This rising trend can partly be attributed to the growing number of commercial vehicle owners converting from petrol engines to gas-powered engines.

Figure 3.4

LPG Production (kt) in Ghana (Anon, 2020a)



CHAPTER FOUR: PROSPECTS AND CHALLENGES OF NATURAL GAS UTILISATION IN GHANA

4.1 Prospects for Natural Gas Utilisation in Ghana

Natural gas accounts for a substantial share of Ghana's total primary energy supply and has met a majority of the country's power needs in recent years. Given this significance, disruption to the gas value chain carries serious consequences for the wider economy, and particularly the power sector. A share of Ghana's gas fields remain undeveloped, due in part to limited field development or storage constraints, suggesting the sector holds meaningful untapped potential to meet rising energy demand if utilisation is accelerated.

4.1.1 Industrial Fuels

As a relatively low-cost, high-quality, and clean energy source, natural gas could be promoted by government as an alternative to coal, supporting boiler conversions and emissions reduction across process heating, combined heat and power, and petrochemical feedstock applications. Ghana's industrialisation strategy has included petrochemical projects that draw on gas feedstock, and continued expansion into LNG, fertilizer, and agrochemical production would point to further growth in utilisation ahead. Industrial gas demand does, however, remain sensitive to the relative pricing of coal, fuel oil, electricity, and other energy alternatives.

4.1.2 Gas for Power Generation

The power sector stands to gain significantly from continued development of Ghana's gas resources. Historically reliant on hydroelectric output, Ghana expanded thermal generation capacity following periods of below-average rainfall combined with rising electricity demand, and gas-fired generation in particular has grown in importance. Expanding gas-fired power generation can help shorten payback periods by increasing utilisation of pipeline transmission capacity, while supporting seasonal peak-shaving and overall supply security.

4.1.3 Transportation Gas

Gas use in Ghana's transport sector has grown steadily, with a meaningful share of LPG supplied nationally consumed by the transport sector, driven by commercial vehicle owners converting engines to run on gas for cost reasons. Given that gas remains cheaper than conventional fuels, further growth in gas-powered vehicles appears likely, though the longer-term emergence of solar and electric vehicles may introduce future competition.

4.2 Challenges Facing Natural Gas Utilisation in Ghana

Despite its potential, Ghana's gas sector faces a range of structural challenges that have limited its growth. Although still a relatively young industry, natural gas could meaningfully contribute to economic growth and sustainable development, provided the sector's underlying challenges around gas gathering and distribution, fiscal and regulatory policy, storage capacity, and investment levels are addressed.

4.2.1 Industry Structure and Regulatory Framework

Opinions differ within the industry on the most suitable structure for Ghana's gas sector. The current model is a state-owned, vertically integrated structure with GNPC overseeing the full chain, from upstream ownership through midstream infrastructure to downstream distribution, supported by the Ministry of Petroleum and the Ghana National Gas Company (GNGC), with GNPC acting as sole owner of domestically produced gas, national gas aggregator, and owner of midstream and downstream infrastructure.

A central concern for a still-developing industry like Ghana's is whether this vertically integrated model can remain viable and continue attracting infrastructure investment, given its ties to an already debt-burdened power sector and the lack of clear division between upstream/downstream and technical/economic regulatory oversight. Vertical integration models typically suit early-stage industries with a single supplier and buyer; with several active supply sources now in place, the industry has arguably moved past that early stage, suggesting the current structure may be overdue for reform. More broadly, vertical integration tends to shift bargaining power toward the integrated firm and can limit innovation by competitors (Riordan, 2008).

GNPC's dual role as both industry regulator and market participant compounds this issue, as the corporation purchases gas from supply streams and sets pricing for the Ghana Power Plant, the transmission pipeline, and final customers, differing from markets where pricing is benchmarked to gas-to-gas competition or other commodity indices. The broader market structure also suffers from limited private-sector participation, contributing to capacity-balancing issues. Abudu and Sai (2020) note that the natural gas pricing policy introduced in 2012 does not allow market forces of supply and demand to operate freely, discouraging further investment.

4.2.2 Gas Transmission and Distribution

Ghana's gas transmission and distribution sector faces both operational and environmental challenges. Operationally, the downstream regulator granted exclusive transportation rights over infrastructure built by GNGC to a separate bulk storage and transport entity, creating ambiguity around operational responsibility between producers, sellers, and gas users, and contributing to institutional friction that has undermined the sector's development.

Hydrate formation within pipeline networks presents a further operational challenge. Hydrates form when gas molecules become trapped within hydrogen-bonded water molecules under low-temperature, high-pressure conditions, solidifying into crystal structures (Broni-Bediako and Amarin, 2017). Depending on volume and duration of exposure, hydrate formation can disrupt fluid flow, ranging from minor operational issues to more serious safety hazards, and can cause pipeline blockages that create hazardous conditions and costly repairs, as has been documented in the Keta basin.

On the environmental side, poor road conditions across much of Ghana hamper the safe transportation of LPG to different regions, and tanker accidents linked to bad roads have at times resulted in loss of life and financial costs for downstream gas companies, continuing to hinder the broader development of Ghana's gas resources.

4.2.3 Gas Storage Problems

Inadequate storage capacity remains a significant barrier to effective gas utilisation in Ghana. Much of the gas produced is either flared or reinjected into wells, with surplus beyond what downstream consumers require via pipeline typically flared rather than stored, representing a direct loss in potential revenue. Flaring is generally reserved for gas deemed uneconomical to capture, yet in Ghana's case, commercially viable volumes have at times been flared due to a lack of adequate storage infrastructure. In the absence of storage options, flaring becomes necessary to manage pressure build-up in associated and non-associated gas wells, prevent leakage from pressure seals, and manage routine pressure releases.

CHAPTER FIVE: CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

Natural gas remains critical to Ghana's economic development, and its effective utilisation could meaningfully address challenges across several sectors of the economy, particularly power generation. Domestic gas production and importation have followed uneven trajectories: while production has generally trended upward, gas imports from Nigeria via WAGP have become less reliable at times, driven partly by feedstock constraints and partly by Ghana's difficulty meeting its debt obligations. Current gas pricing policy also appears to allow GNPC to generate substantial margins through its pricing structure, pointing to a need for pricing policy reform to improve overall market functioning.

Meanwhile, natural gas utilisation in Ghana has benefited from steady progress in pipeline distribution and transmission infrastructure, improvements in the balance of gas supply and demand, and continued growth in LPG usage. That said, the full economic potential of natural gas remains constrained by structural and regulatory challenges, a largely state-controlled monopoly structure that limits private-sector participation and investment, ongoing operational and environmental difficulties in gas transmission and distribution, and inadequate storage infrastructure that results in usable gas being flared.

Nonetheless, natural gas continues to hold significant potential for Ghana's economy, whether as an industrial fuel, a power generation resource, or a transportation fuel, and could meaningfully strengthen the national economy if fully developed and responsibly managed.

5.2 Recommendations

Based on the above, the following recommendations are proposed:

1. Increased investment in gas distribution infrastructure, covering gathering, transmission, and distribution, is needed to strengthen the full gas value chain and support the broader development of natural gas utilisation in Ghana.
2. Government should create a more enabling environment for private-sector participation in gas development, alongside improvements in social infrastructure and capacity building across the sector.
3. Given the current structural challenges in Ghana's gas industry, it would be more appropriate to unbundle midstream and downstream infrastructure ownership from natural gas supply, with GNPC preserving its aggregating role while infrastructure ownership is separated from upstream producers.

REFERENCES

- Abudu, H. and Sai, R. (2020). Examining Prospects and Challenges of Ghana's Petroleum Industry: A Systematic Review. *Energy Reports*, 6, 841-858.
- Adam, M. A. (2014). Three Years of Petroleum Revenue Management in Ghana: Transparency Without Accountability. Africa Centre for Energy Policy (ACEP) Policy Brief.
- Agbonifio, P. E. (2013). Opportunities, Challenges and Obstacles to Economic Growth and Sustainable Development through Natural Gas. *Sustainable Development in Africa*.
- Amorin, R. (2018). The Use of Liquefied Petroleum Gas (LPG) as a Fuel for Commercial Vehicles in Ghana: A Case Study at Tema Community 1. *Current Journal of Applied Sciences and Technology*, 29(2), 1-8.
- Amponsah, K. and Opei, N. (2014). Downstream Petroleum Industry in Ghana: Challenges and Opportunities.
- Anon. (2008). Uses of Natural Gas. U.S. Energy Information Administration. <https://www.eia.gov/energyexplained>
- Anon. (2010). Ghana Gas Activities. Ghana National Gas Company. <https://www.ghanagas.com.gh>
- Anon. (2012). Natural Gas Pricing Policy for Ghana. <https://www.reportingoilgas.org>
- Anon. (2013). Energizing Economic Growth in Ghana: Making the Power and Petroleum Sector Rise to the Challenges. World Bank. <https://openknowledge.worldbank.org/handle/10986/16264>
- Anon. (2015). Issues with Gas Flaring. U.S. Energy Information Administration. <https://www.eia.gov/energyexplained>
- Anon. (2016). Petroleum Sector: Energy Supply and Demand Outlook for Ghana. Energy Commission of Ghana. <https://www.energycom.gov.gh>
- Anon. (2017a). Ghana Natural Gas Reserves; Production and Consumption. Reuters. <https://www.reuters.com>
- Anon. (2017b). Physiochemical Characterization of the Jubilee Crude Oil. <https://www.elixirpublishers.com>
- Anon. (2019). Petroleum Sector Supply and Demand Outlook. Energy Commission of Ghana.
- Anon. (2020a). Ghana Gas Activities and Infrastructure Report. Ghana National Gas Company.
- Anon. (2020b). Natural Gas Production and Import Statistics. Ghana National Petroleum Corporation.
- Broni-Bediako, E. and Amorin, R. (2017). Gas Hydrate Formation Phase Boundary Behaviour of Synthetic Natural Gas System of the Keta Basin of Ghana. *The Open Petroleum Engineering Journal*, 10(1), 304-338.

Fulwood, M. and Bros, T. (2018). Future Prospects for LNG Demand in Ghana. Oxford Institute for Energy Studies, Energy Insight: 26.

Obanijesu, E. O. and Sonibare, J. A. (2005). Natural Gas Flaring and Environmental Protection in Nigeria. *NAFTA Journal*, 65-108.

Obanijesu, E. O. (2009). West Africa Gas Pipeline Project: Associated Problems and Possible Remedies (Published MSc Thesis). Universiti Teknologi Petronas, Malaysia.

Occhiali, G. and Hafner, M. (2017). Ghanaian Energy Markets and the Role of Natural Gas for Local Development. Fondazione Eni Enrico Mattei (FEEM), Reports, December.

Patin, S. (2004). *Gas Impact on Marine Organisms*.

Riordan, M. H. (2008). Competitive Effects of Vertical Integration. In *Handbook of Antitrust Economics*.

Sulemana, S. (2020). Natural Gas Industry Restructuring for Value Optimization: A Case Study of Ghana. *International Journal of Strategic Energy and Environmental Planning*, 2(6).